

Rhombix Technologies

DevOps Internship — Task Submission

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Task Selected	Project 2: Containerization with Docker
Organization	Rhombix Technologies

1. Task Overview

This report documents the completion of Project 2 — Containerization with Docker — assigned as part of the Rhombix Technologies DevOps internship. The objective of this task was to gain hands-on experience with Docker, learning how to package, build, run, and manage containerized applications.

The task involved pulling images from Docker Hub, writing a custom Dockerfile, building a Docker image, running containers with port mapping, and managing container lifecycle operations such as starting, stopping, and monitoring logs.

Key Learning Objectives

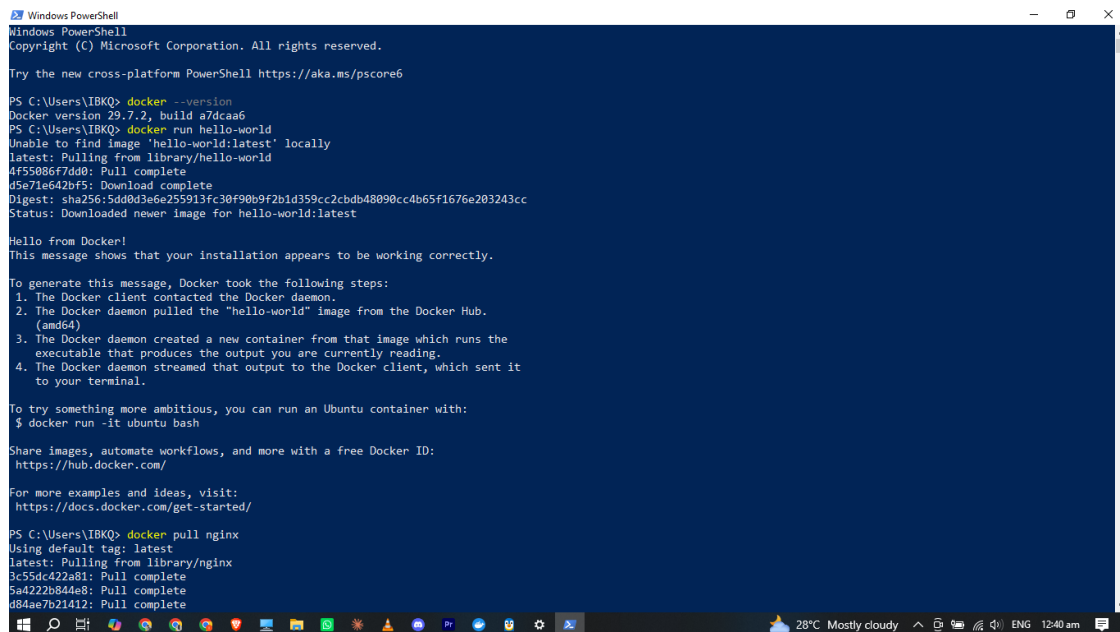
- Verifying Docker installation and running a test container using `docker run hello-world`.
- Pulling pre-built images from Docker Hub, such as the official nginx image.
- Running a container from an image and mapping container ports to host ports.
- Writing a custom Dockerfile to build a personalized web image.
- Building a Docker image from a Dockerfile using `docker build`.
- Running multiple containers simultaneously and verifying them in Docker Desktop.
- Managing container lifecycle using `docker ps`, `docker stop`, and `docker start`.
- Viewing container logs using `docker logs` to confirm the service is running correctly.

2. Task Execution — Screenshots

Below are the screenshots captured while performing the assigned Docker containerization task.

2.1 Docker Installation Check & Running First Container

The Docker version was verified using `docker --version`, followed by running the hello-world image to confirm that the Docker installation was working correctly. The nginx image was then pulled from Docker Hub.



```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\IBKQ> docker --version
Docker version 29.7.2, build a7dca6
PS C:\Users\IBKQ> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55886f7d40: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e25913fc30f90b9f2b1d359cc2cddb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

PS C:\Users\IBKQ> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
3c55dc422a81: Pull complete
5a4222b84ae8: Pull complete
d84ae7b21412: Pull complete
```

Fig 1: Verifying Docker installation, running hello-world container, and pulling the nginx image

2.2 Running the Nginx Container

The official nginx image was run as a container and mapped to port 8080 on the host machine. Accessing localhost:8080 in the browser confirmed that the nginx web server was successfully running inside the container.

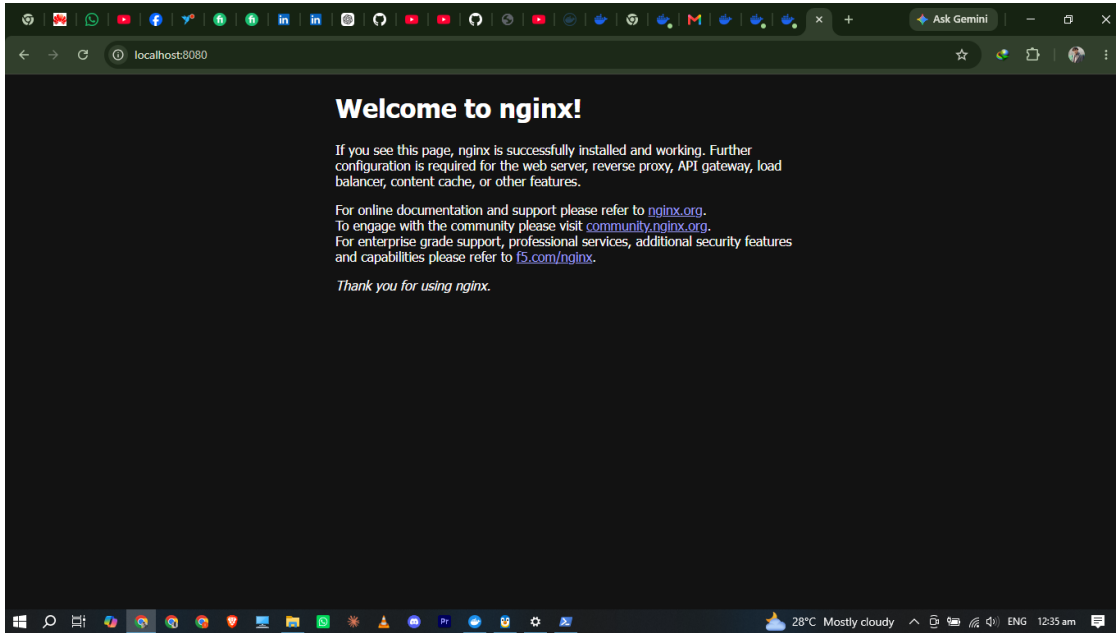


Fig 2: Default nginx welcome page accessed via localhost:8080, confirming the container is running

2.3 Building a Custom Docker Image

A custom Dockerfile was created, based on the nginx image, which copies a custom index.html file into the container. The docker build command was used to build this image, successfully creating the rhombix-web image.

```
Windows PowerShell
[+] Building 1.8s (7/7) FINISHED
    docker:desktop-linux
=> [internal] load build definition from Dockerfile
    0.0s
=> => transferring dockerfile: 119B
    0.0s
=> [internal] load metadata for docker.io/library/nginx:latest
    0.2s
=> [internal] load .dockerignore
    0.1s
=> => transferring context: 2B
    0.0s
=> [internal] load build context
    0.1s
=> => transferring context: 321B
    0.0s
=> [1/2] FROM docker.io/library/nginx:latest@sha256:8541484afbc9c8a5a8a99b379568ebbc957f658583ec9448fc43104229c03cf8
    0.4s
=> => resolve docker.io/library/nginx:latest@sha256:8541484afbc9c8a5a8a99b379568ebbc957f658583ec9448fc43104229c03cf8
    0.1s
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
    0.1s
=> exporting to image
    0.6s
=> => exporting layers
    0.2s
=> => exporting manifest sha256:5fba4965d72ad14d7d20f4f284bad0d851b5bf1da6e7854ad6a057c82a5bbb9
    0.0s
=> => exporting config sha256:0d666eb4c059c1777e3a0c41e81a246bdaab32def0bda8f04ec1b4449f27a5b8
    0.0s
=> => exporting attestation manifest sha256:ba006f46763d40b36b4e9cca931d2a23a4707fd07a36026909bcacb2822697b
    0.1s
=> => exporting manifest list sha256:e3afd5bb7172dd785a4858ca55c6a1ecc0f4b72cde6f420943e33fad2bbc4767
    0.0s
=> => naming to docker.io/library/rhombix-web:latest
```

Fig 3: Building a custom Docker image (rhombix-web) from a Dockerfile

2.4 Verifying Images in Docker Desktop

Docker Desktop was used to verify that all images — hello-world, nginx, and the custom ibkq/rhombix-web — were successfully created and available locally.

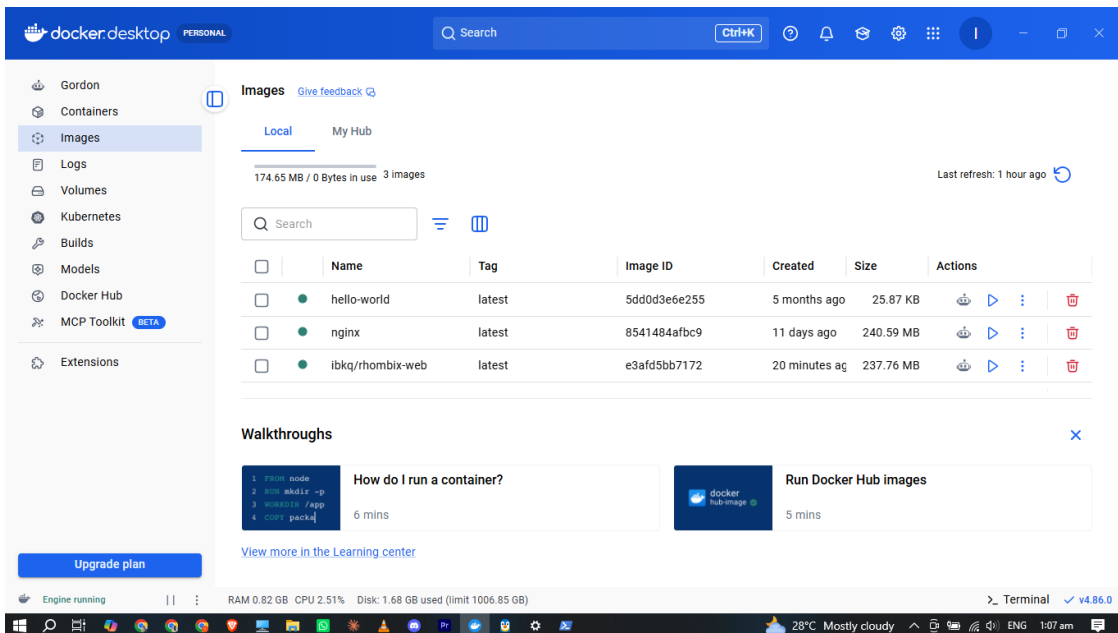


Fig 4: Docker Desktop showing the local images: hello-world, nginx, and custom rhombix-web image

2.5 Running and Managing Multiple Containers

A container was launched from the custom rhombix-web image and mapped to port 8081, running alongside the existing nginx container on port 8080. The docker ps command was used to confirm both containers were running, and docker stop / docker start were used to test container lifecycle management.

```

CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
ibkq-rhombix-container
978e8d6c3b13   ibkq/rhombix-web:latest            "/docker-entrypoint..." 14 seconds ago Up 13 seconds 0.0.0.0:8081->80/tcp, [::]:8081->80
/tcp
4f6dc42cb2d0   nginx                              "/docker-entrypoint..." 20 minutes ago Up 20 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80
/tcp
rhombix-nginx
PS C:\Users\IBKQ\rhombix-docker-task> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
ibkq-rhombix-container
978e8d6c3b13   ibkq/rhombix-web:latest            "/docker-entrypoint..." 3 minutes ago  Up 3 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80
/tcp
4f6dc42cb2d0   nginx                              "/docker-entrypoint..." 23 minutes ago Up 23 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80
/tcp
rhombix-nginx
PS C:\Users\IBKQ\rhombix-docker-task> docker stop ibkq-rhombix-container
ibkq-rhombix-container
PS C:\Users\IBKQ\rhombix-docker-task> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
4f6dc42cb2d0   nginx                              "/docker-entrypoint..." 25 minutes ago Up 25 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
rhombix-nginx
PS C:\Users\IBKQ\rhombix-docker-task> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
978e8d6c3b13   ibkq/rhombix-web:latest            "/docker-entrypoint..." 5 minutes ago   Exited (0) 23 seconds ago
ibkq-rhombix-container
4f6dc42cb2d0   nginx                              "/docker-entrypoint..." 25 minutes ago Up 25 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
rhombix-nginx
80fe83aaf0ca   hello-world                       "/hello"           31 minutes ago Exited (0) 31 minutes ago
dreamy_goldwasser
PS C:\Users\IBKQ\rhombix-docker-task> docker start ibkq-rhombix-container
ibkq-rhombix-container
PS C:\Users\IBKQ\rhombix-docker-task> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
978e8d6c3b13   ibkq/rhombix-web:latest            "/docker-entrypoint..." 6 minutes ago   Up 9 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80
/tcp
ibkq-rhombix-container

```

Fig 5: Running two containers simultaneously and testing docker stop / docker start commands

2.6 Container Logs and Status Verification

The docker logs command was used to view the nginx entrypoint logs of the custom container, confirming that the web server configuration loaded and started correctly inside the container.

```

2026/08/15 19:59:32 [notice] 1#1: start worker process 22
2026/08/15 19:59:32 [notice] 1#1: start worker process 23
2026/08/15 19:59:32 [notice] 1#1: start worker process 24
2026/08/15 19:59:32 [notice] 1#1: start worker process 25
PS C:\Users\IBKQ\rhombix-docker-task> docker stop ibkq-rhombix-container
>> docker ps
>> docker ps -a
>> docker start ibkq-rhombix-container
>> docker ps
ibkq-rhombix-container
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
4f6dc42cb2d0   nginx                              "/docker-entrypoint..." 29 minutes ago Up 29 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
rhombix-nginx
CONTAINER ID   IMAGE                                NAMES                    COMMAND                  CREATED        STATUS        PORTS
978e8d6c3b13   ibkq/rhombix-web:latest            ibkq-rhombix-container  "/docker-entrypoint..." 8 minutes ago   Exited (0) Less than a second ago
4f6dc42cb2d0   nginx                              rhombix-nginx          "/docker-entrypoint..." 29 minutes ago Up 29 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
80fe83aaf0ca   hello-world                       "/hello"               35 minutes ago Exited (0) 35 minutes ago
dreamy_goldwasser
ibkq-rhombix-container
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
978e8d6c3b13   ibkq/rhombix-web:latest            "/docker-entrypoint..." 8 minutes ago   Up Less than a second  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
4f6dc42cb2d0   nginx                              rhombix-nginx          "/docker-entrypoint..." 29 minutes ago Up 29 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
PS C:\Users\IBKQ\rhombix-docker-task> docker logs ibkq-rhombix-container
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh

```

Fig 6: Verifying container status with docker ps -a and viewing container logs with docker logs

2.7 Final Output — Custom Web Page in Container

The custom container was accessed via localhost:8081, displaying the personalized webpage confirming that the custom Docker image was built and deployed successfully as part of the DevOps Internship Task 2.

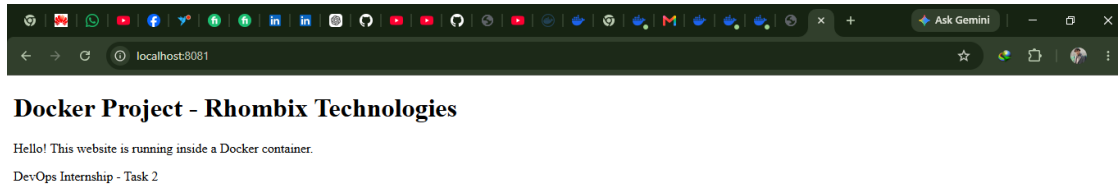


Fig 7: Custom webpage 'Docker Project - Rhombix Technologies' served from the container on localhost:8081

3. Conclusion

Through this task, practical experience was gained in containerizing applications using Docker. Pulling images, writing a custom Dockerfile, building images, running multiple containers with port mapping, and managing container lifecycle operations provided a strong foundation in containerization — a core skill in modern DevOps workflows.

This task has been completed as part of the Rhombix Technologies DevOps Internship Program and is being submitted in accordance with the internship submission guidelines.